

14. | Write a code using RAW sockets to implement packet sniffing.

AIM: To write a Python program using RAW sockets to implement packet sniffing, capturing IP packets and displaying their source IP address and packet size.

Procedure

1. Open a terminal or editor **as Administrator**, because raw sockets need elevated privileges.
2. Get the local IP address with `socket.gethostname(socket.gethostname())`.
3. Create a raw socket with `socket.socket(AF_INET, SOCK_RAW, IPPROTO_IP)`, bind it to the host address and set the `IP_HDRINCL` option so that the IP header is included.
4. Run `python packet_sniffer.py`. In a loop, receive packets with `recvfrom(65535)` and print the source IP address and the packet size.
5. Press Ctrl+C to stop the sniffer; the socket is closed in the *finally* block.

packet_sniffer.py

```
import socket

HOST = socket.gethostname(socket.gethostname())

sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
sniffer.bind((HOST, 0))
sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)

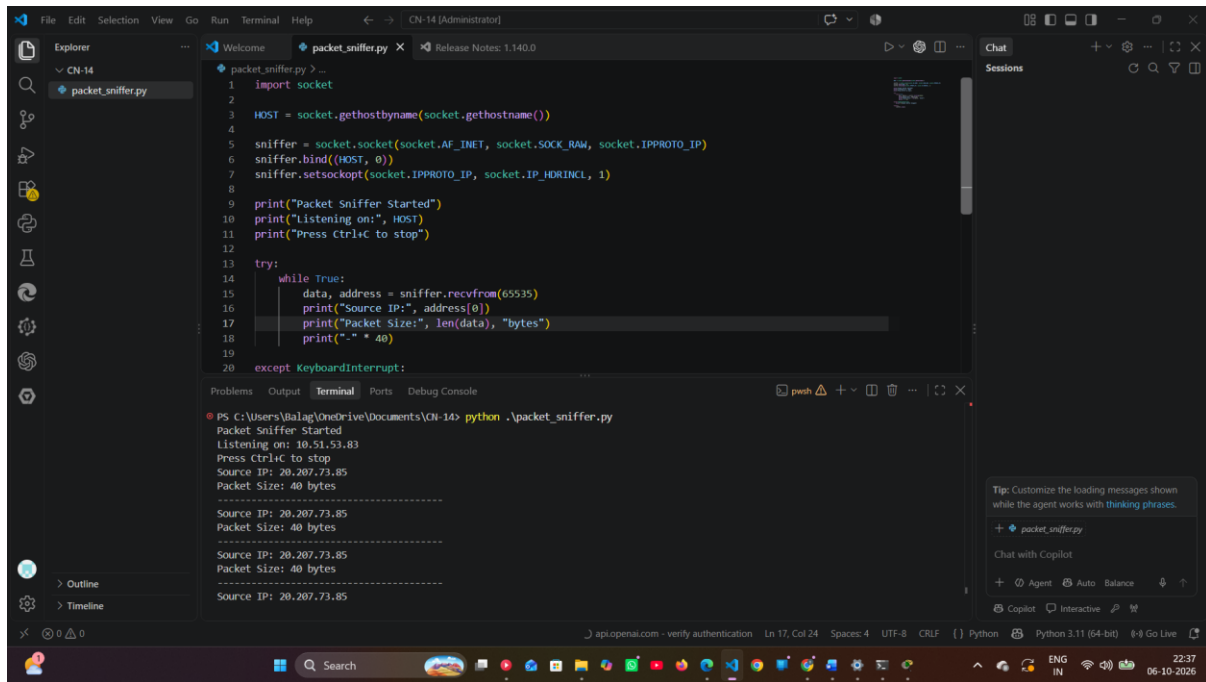
print("Packet Sniffer Started")
print("Listening on:", HOST)
print("Press Ctrl+C to stop")

try:
    while True:
        data, address = sniffer.recvfrom(65535)
        print("Source IP:", address[0])
        print("Packet Size:", len(data), "bytes")
        print("-" * 40)

except KeyboardInterrupt:
    print("\nPacket Sniffer Stopped")

finally:
```

sniffer.close()

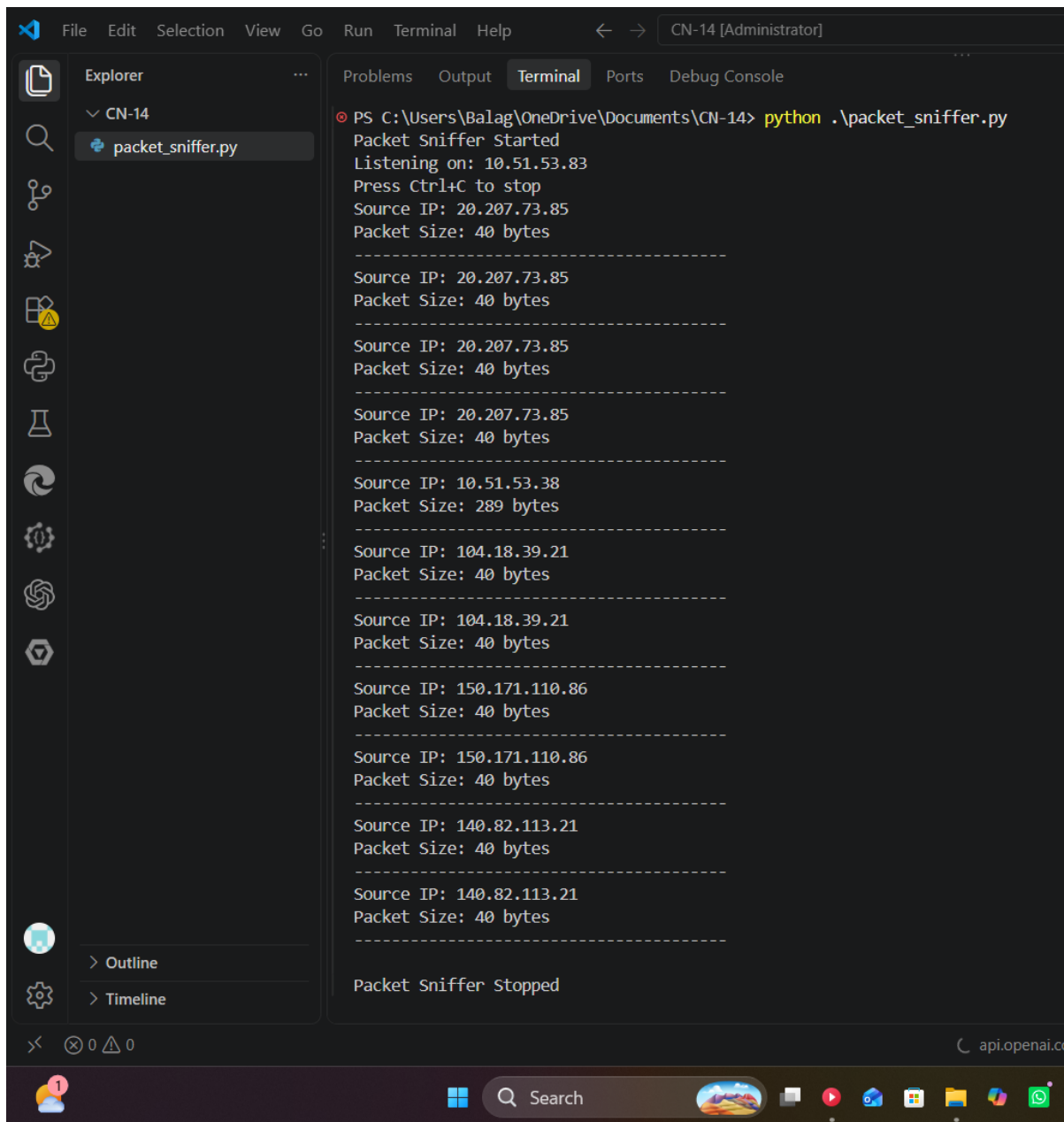


The image shows a Visual Studio Code editor window with a Python script named `packet_sniffer.py` open. The script is a packet sniffer that listens on a specific host and port, receiving data from a remote source. The terminal output shows the script running successfully, displaying the source IP and packet size for each received packet.

```
1 import socket
2
3 HOST = socket.gethostname(socket.gethostname())
4
5 sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
6 sniffer.bind((HOST, 0))
7 sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
8
9 print("Packet Sniffer Started")
10 print("Listening on:", HOST)
11 print("Press Ctrl+C to stop")
12
13 try:
14     while True:
15         data, address = sniffer.recvfrom(65535)
16         print("Source IP:", address[0])
17         print("Packet Size:", len(data), "bytes")
18         print("-" * 40)
19
20 except KeyboardInterrupt:
```

Terminal Output:

```
PS C:\Users\Balag\OneDrive\Documents\CN-14> python .\packet_sniffer.py
Packet Sniffer Started
Listening on: 18.51.53.83
Press Ctrl+C to stop
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
```



The screenshot shows a Windows terminal window titled "CN-14 [Administrator]". The terminal output displays the execution of a Python script named "packet_sniffer.py". The script starts by listening on IP address 10.51.53.83. It then captures several packets from various source IP addresses, displaying the source IP and packet size for each. The packets are separated by dashed lines. The source IPs and packet sizes are: 20.207.73.85 (40 bytes), 10.51.53.38 (289 bytes), 104.18.39.21 (40 bytes), 150.171.110.86 (40 bytes), and 140.82.113.21 (40 bytes). The terminal also shows the command prompt and the file explorer on the left side of the IDE.

```
PS C:\Users\Balag\OneDrive\Documents\CN-14> python .\packet_sniffer.py
Packet Sniffer Started
Listening on: 10.51.53.83
Press Ctrl+C to stop
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 10.51.53.38
Packet Size: 289 bytes
-----
Source IP: 104.18.39.21
Packet Size: 40 bytes
-----
Source IP: 104.18.39.21
Packet Size: 40 bytes
-----
Source IP: 150.171.110.86
Packet Size: 40 bytes
-----
Source IP: 150.171.110.86
Packet Size: 40 bytes
-----
Source IP: 140.82.113.21
Packet Size: 40 bytes
-----
Source IP: 140.82.113.21
Packet Size: 40 bytes
-----
Packet Sniffer Stopped
```

Result

A packet sniffer was implemented in Python using a raw socket. The program listened on 10.51.53.83 and captured IP packets, displaying the source IP address and size of each packet, for example 20.207.73.85 (40 bytes), 10.51.53.38 (289 bytes), 104.18.39.21, 150.171.110.86 and 140.82.113.21 (40 bytes each). The sniffer stopped cleanly when Ctrl+C was pressed.